

生态位模型的原理与应用（二）

THE PRINCIPLES AND APPLICATIONS OF ECOLOGICAL NICHE MODEL

中科院动物所 乔慧捷

2026-9

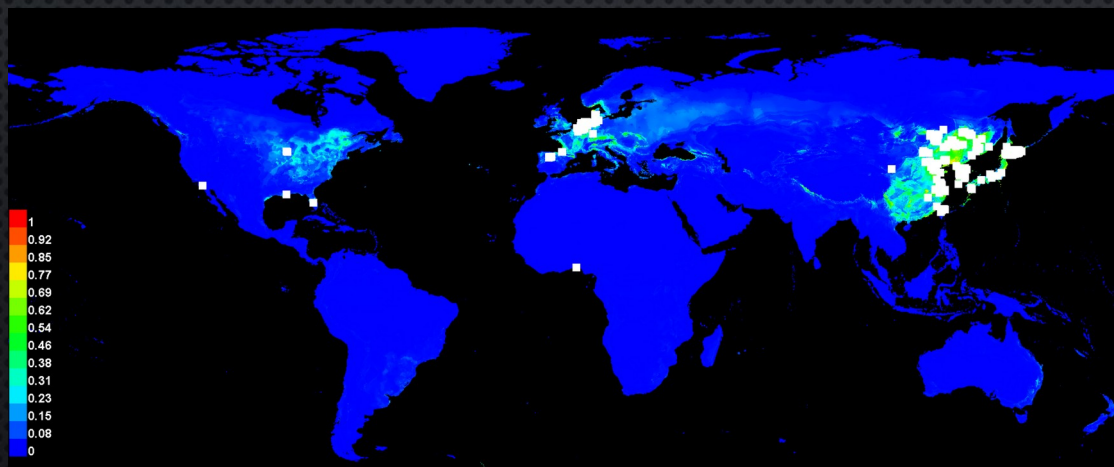
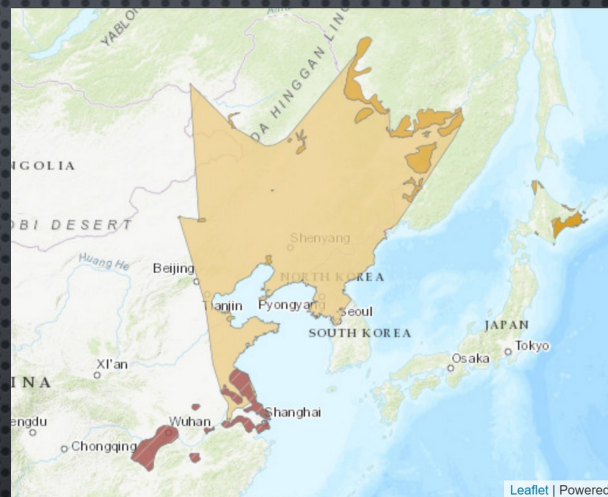
生态位建模方法

- 使用原始发生数据
- 使用栅格环境数据集
- 在模型中结合这两类数据流
- 使用该模型制作地图
- 地图被解释为物种“存在概率”

很简单，对不？

显然，没那么简单！

我们重新回顾一下前面做的事情



过滤物种发生数据

Red-crowned Crane

Grus japonensis has most recently been assessed for *The IUCN Red List of Threatened Species*. *Grus japonensis* is listed as Vulnerable under criteria A2ac+4ac; C1.

Assessed in 2022). *The IUCN Red List of Threatened Species*...

Download ▾

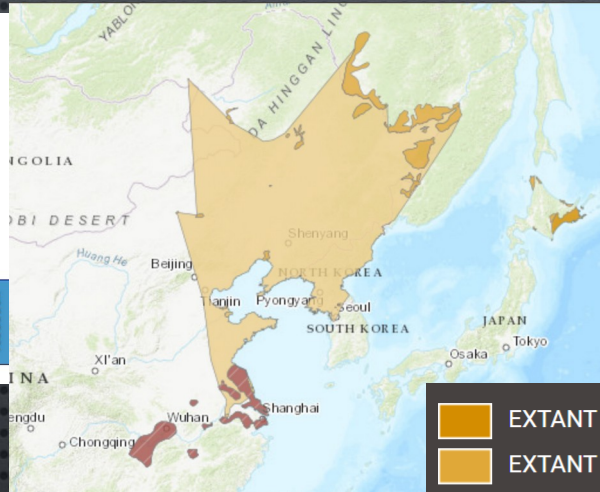
Text Overview

AVAILABLE FILES

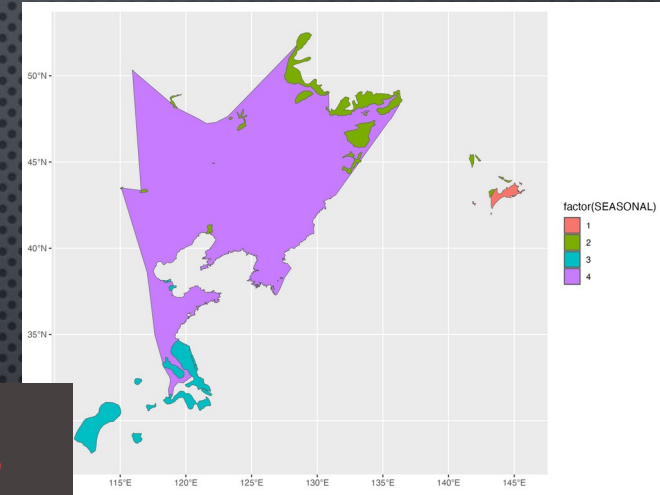
Range data - Polygons (SHP)

Range map (JPG)

feedback



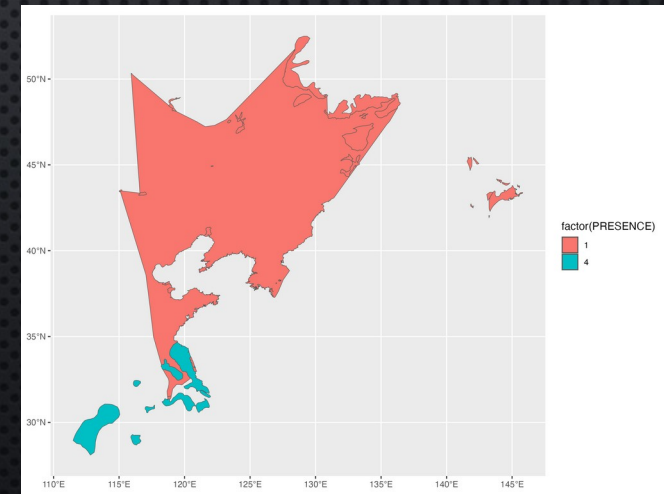
- EXTANT (RESIDENT)
- EXTANT (BREEDING)
- EXTANT (NON-BREEDING)
- EXTANT (PASSAGE)
- POSSIBLY EXTINCT



```
iucn<-read_sf("Red crowned Crane/data_0.shp")
```

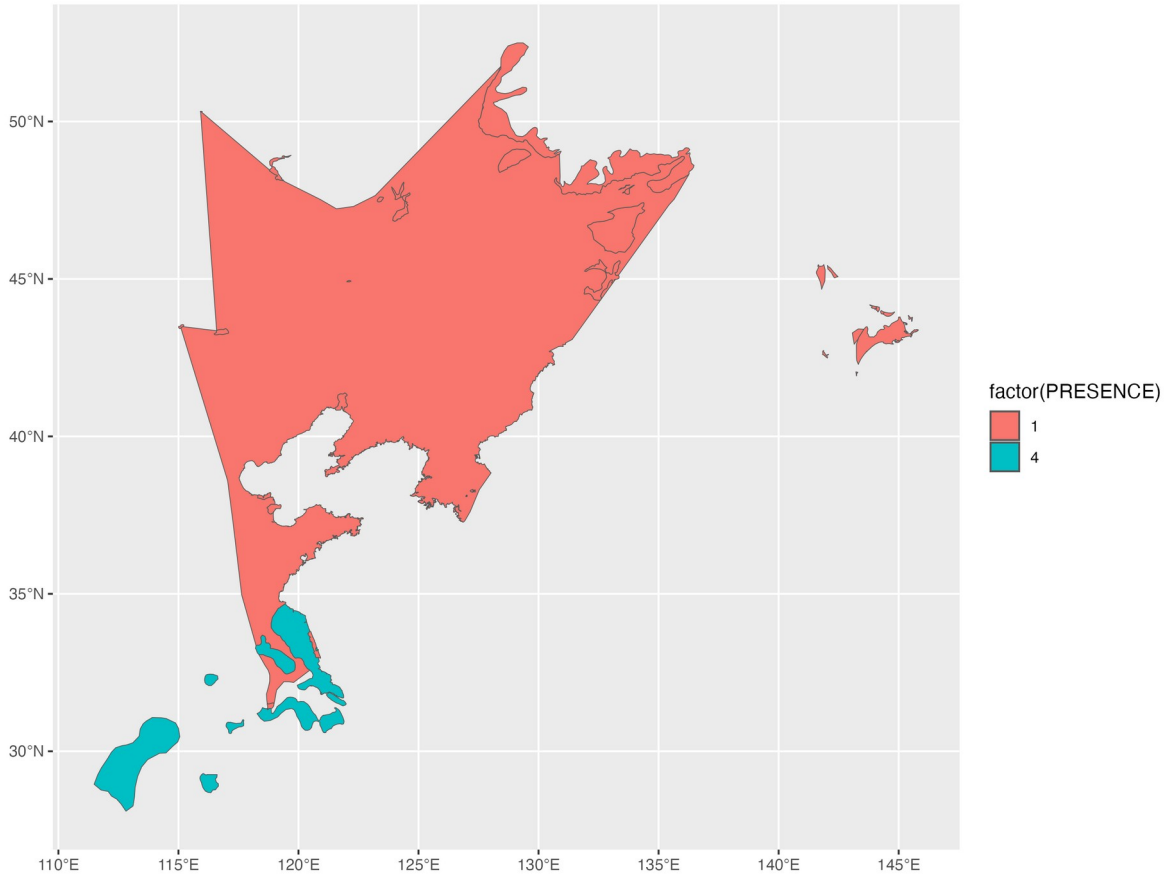
```
ggsave(ggplot(iucn)+  
  geom_sf(aes(fill=factor(PRESENCE))),  
  filename="Figures/IUCN_PRESENCE.png")
```

```
ggsave(ggplot(iucn)+  
  geom_sf(aes(fill=factor(SEASONAL))),  
  filename="Figures/IUCN_SEASONAL.png")
```

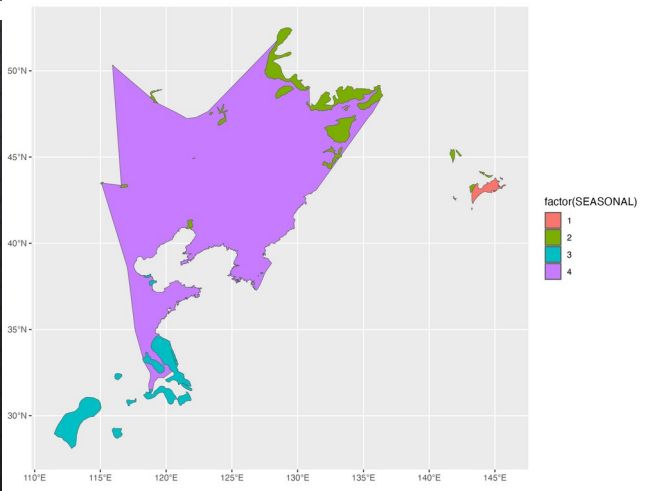


过滤物种发生数据

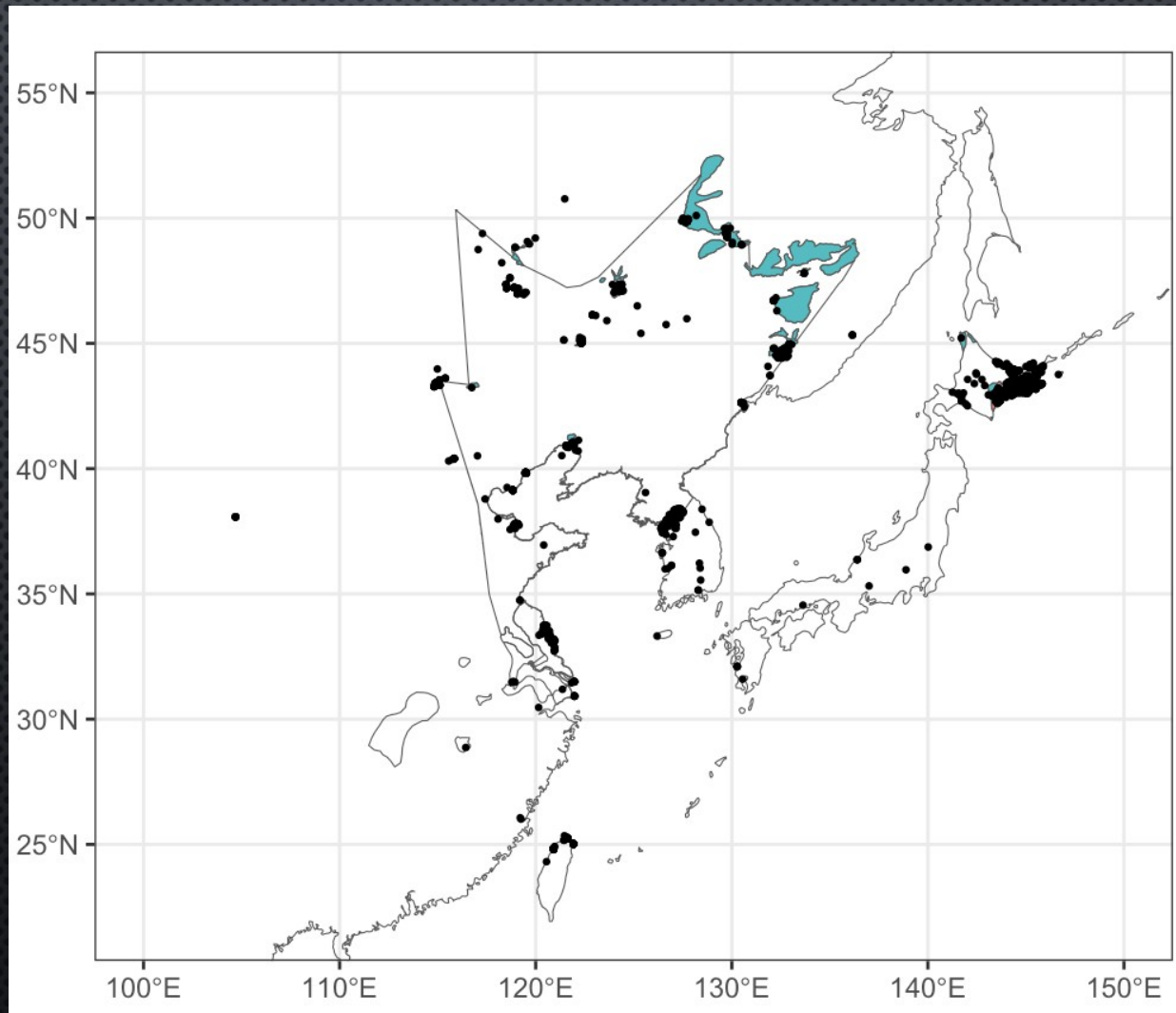
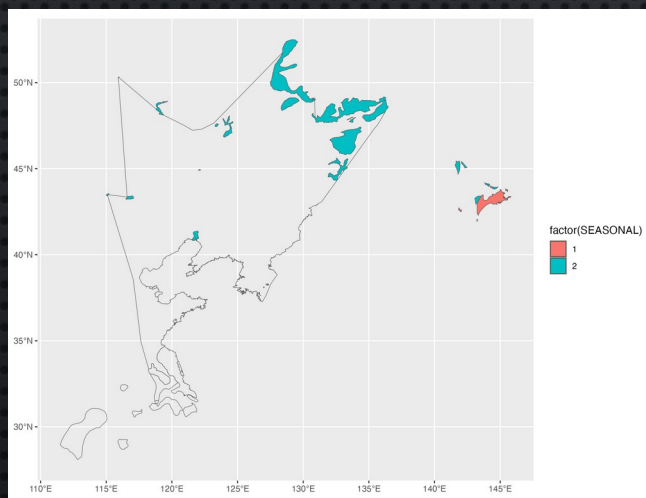
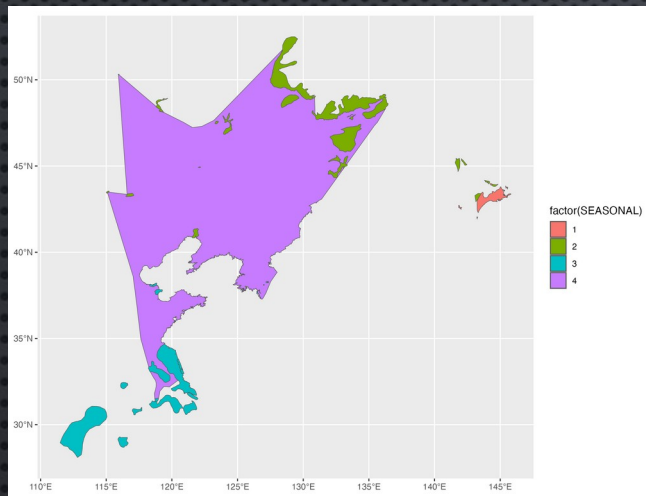
CODE	PRESENCE
1	Extant
2	Probably Extant
3	Possibly Extant
4	Possibly Extinct
5	Extinct
6	Presence Uncertain



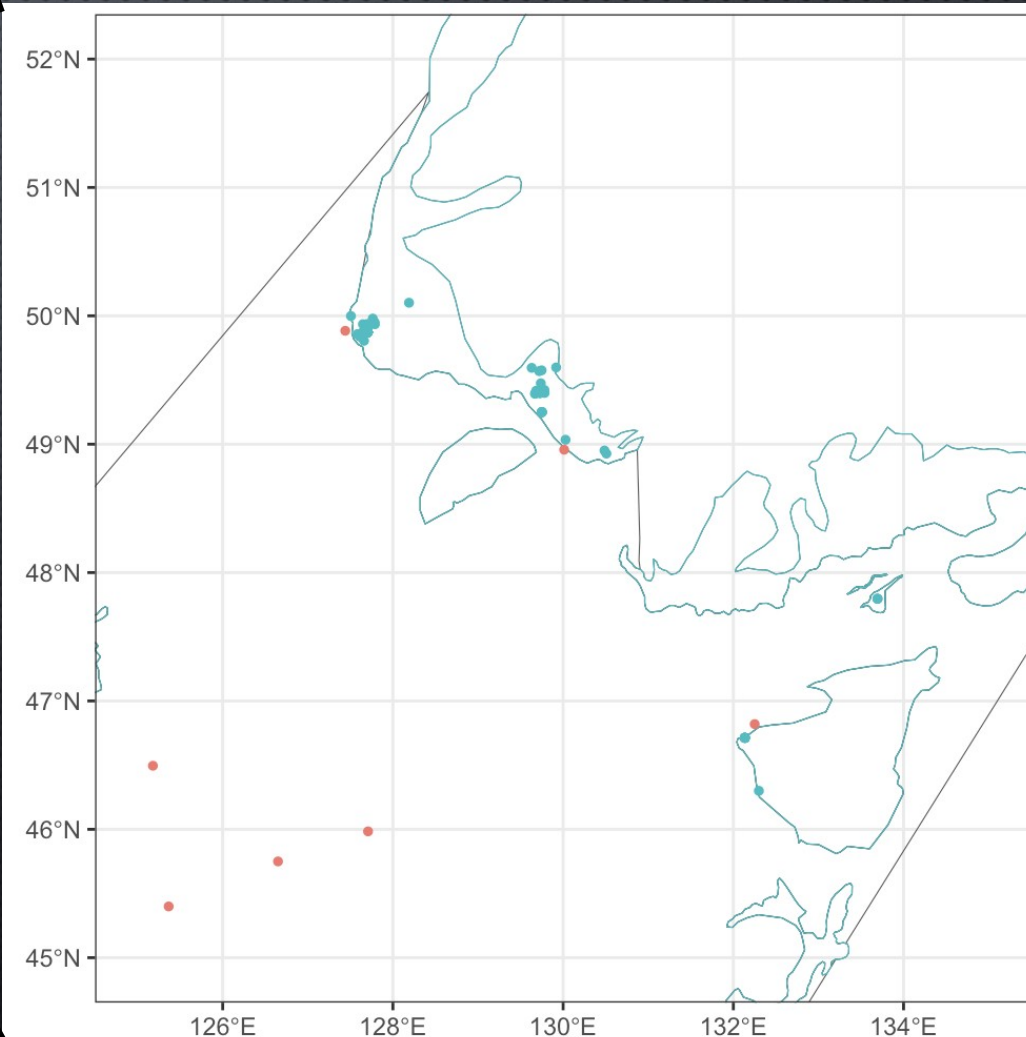
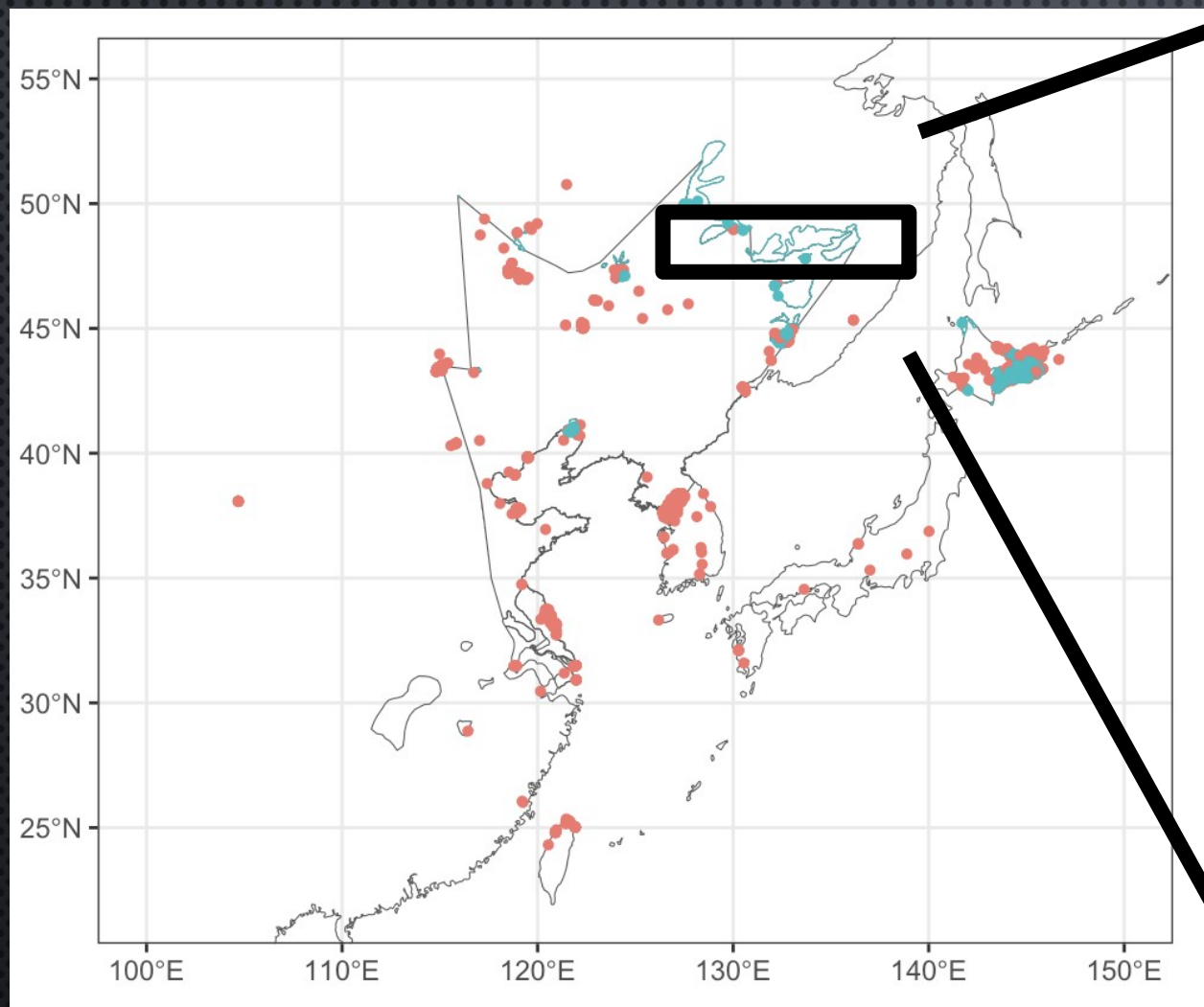
CODE	SEASONALITY
1	Resident
2	Breeding Season
3	Non-breeding Season
4	Passage
5	Seasonal Occurrence Uncertain



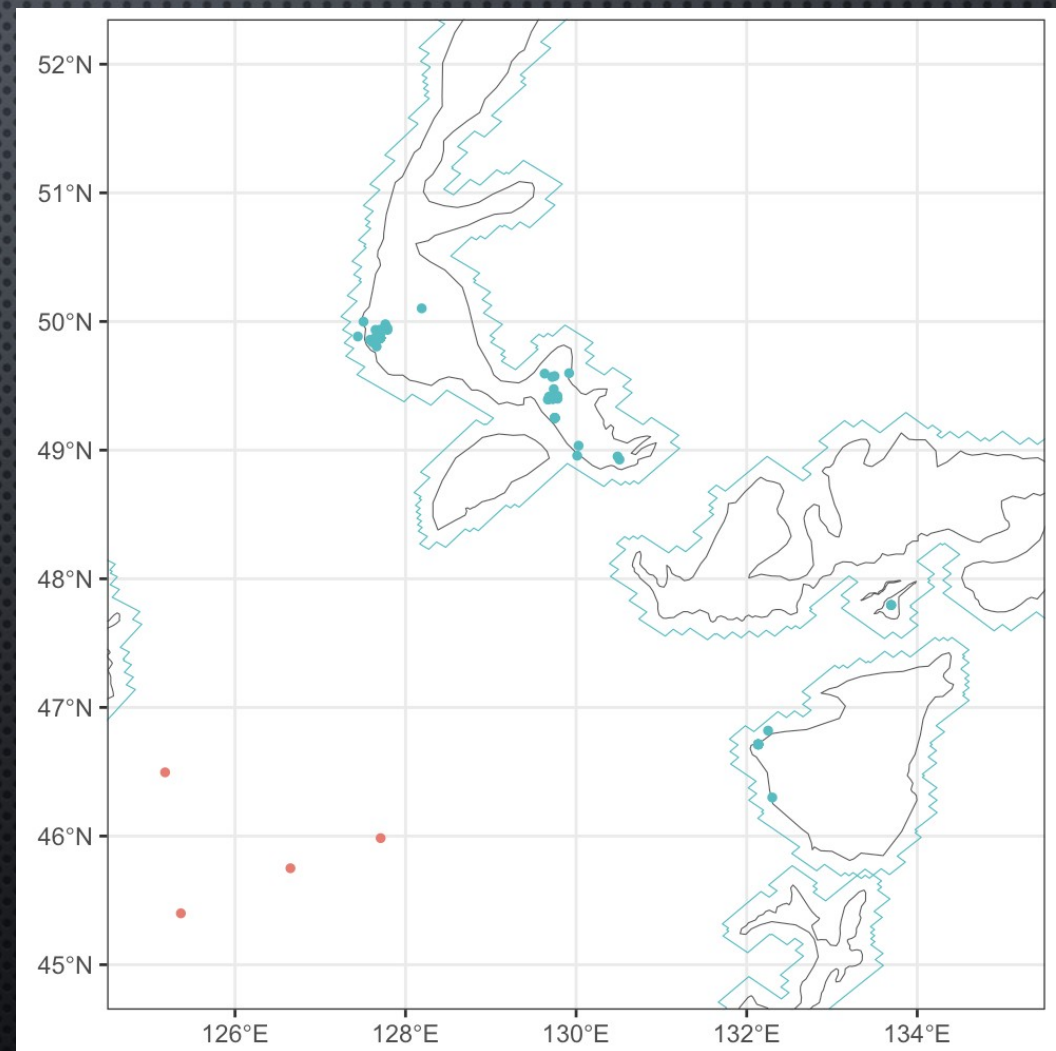
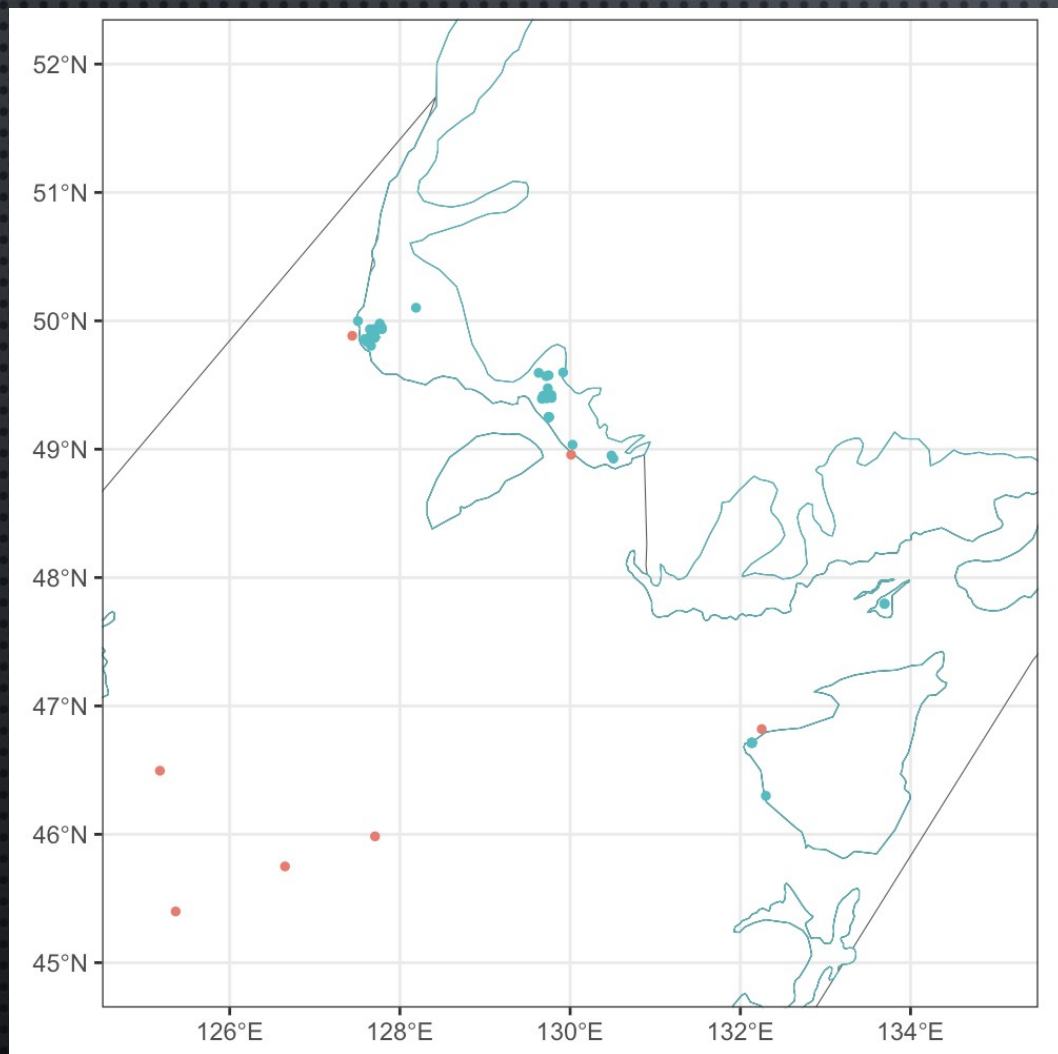
过滤物种发生数据



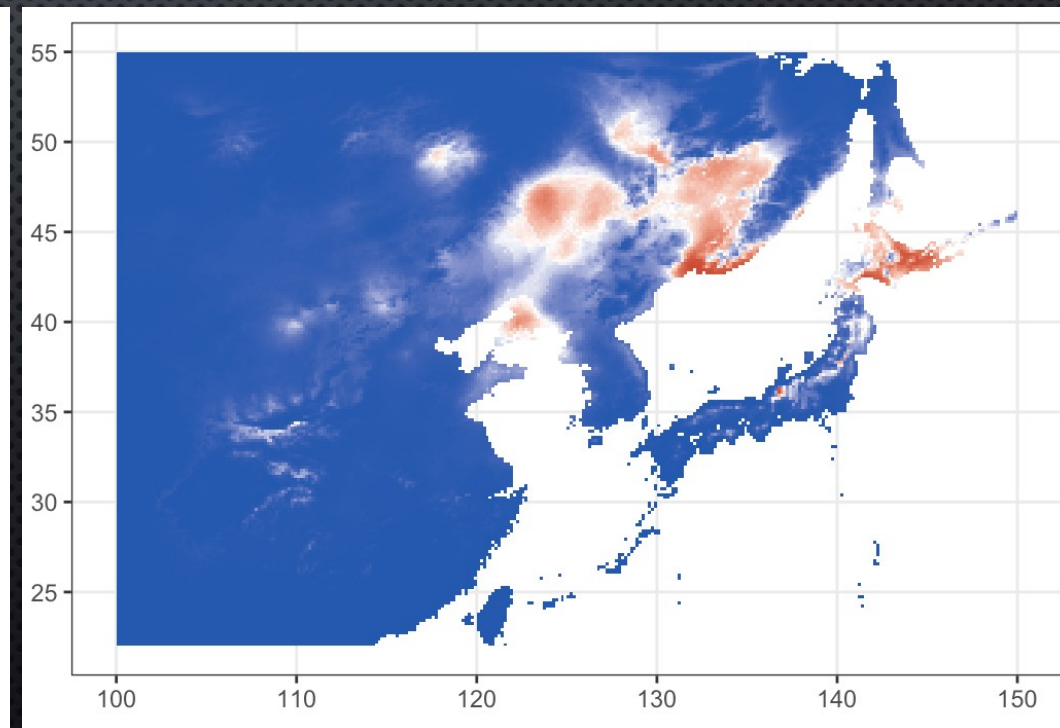
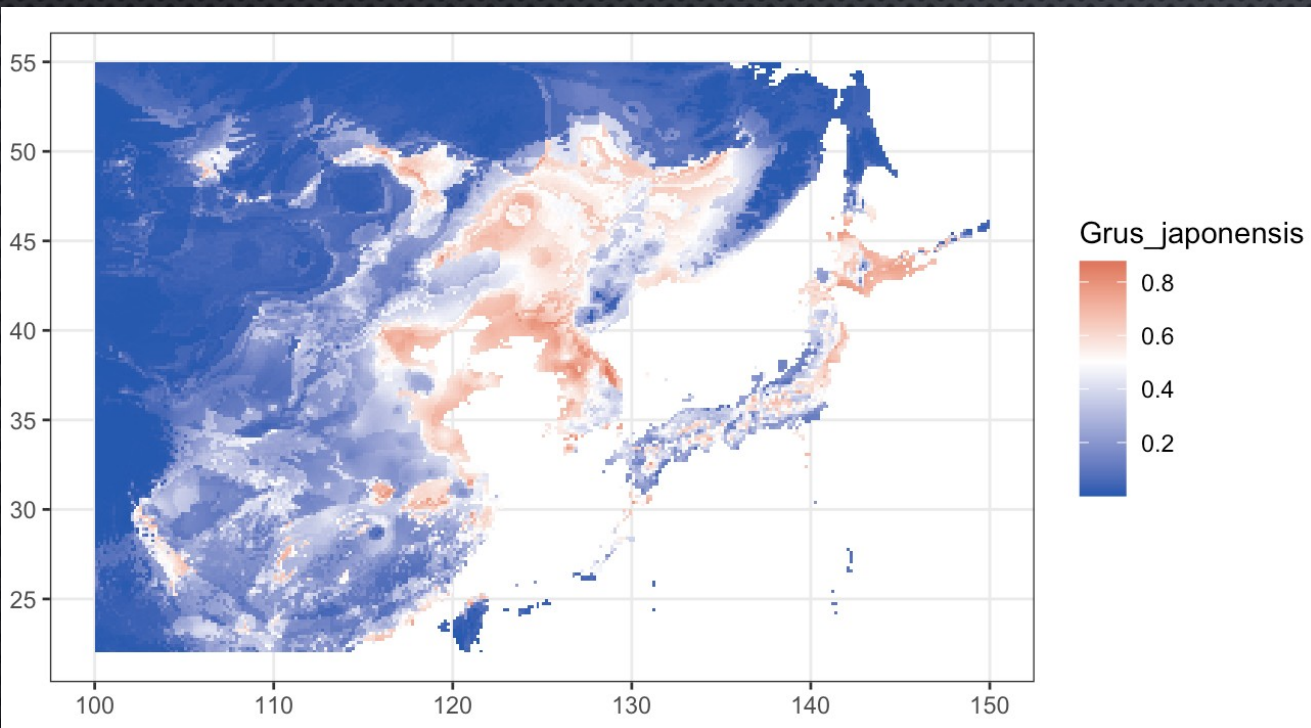
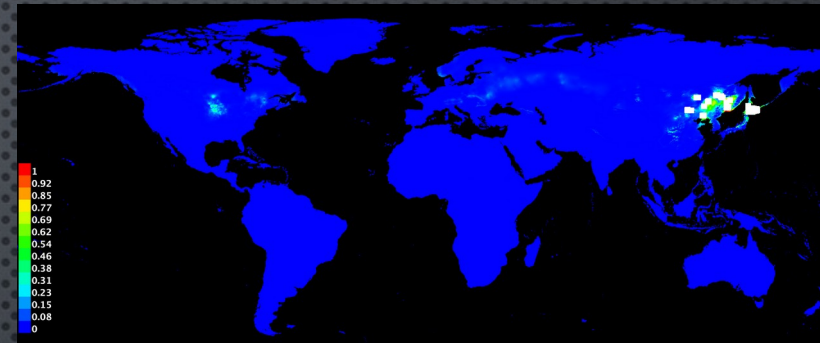
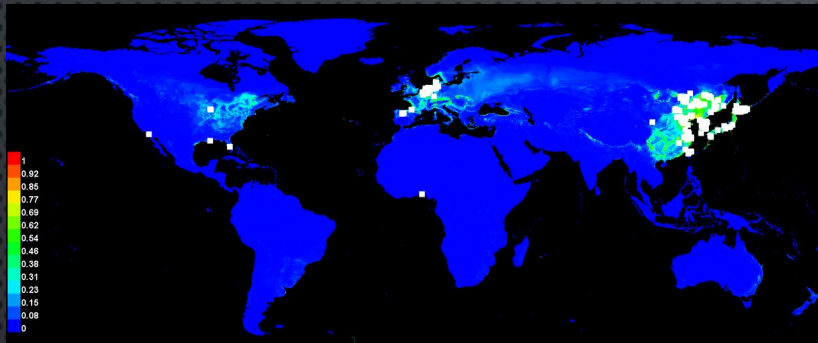
过滤物种发生数据



过滤物种发生数据



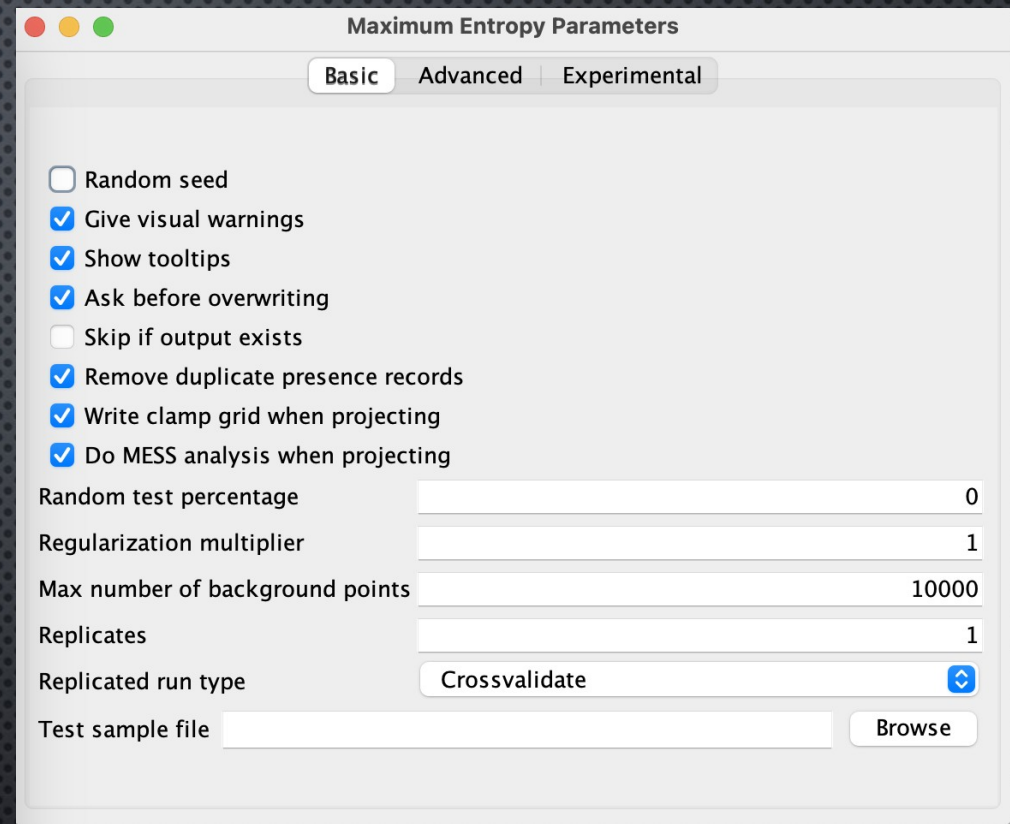
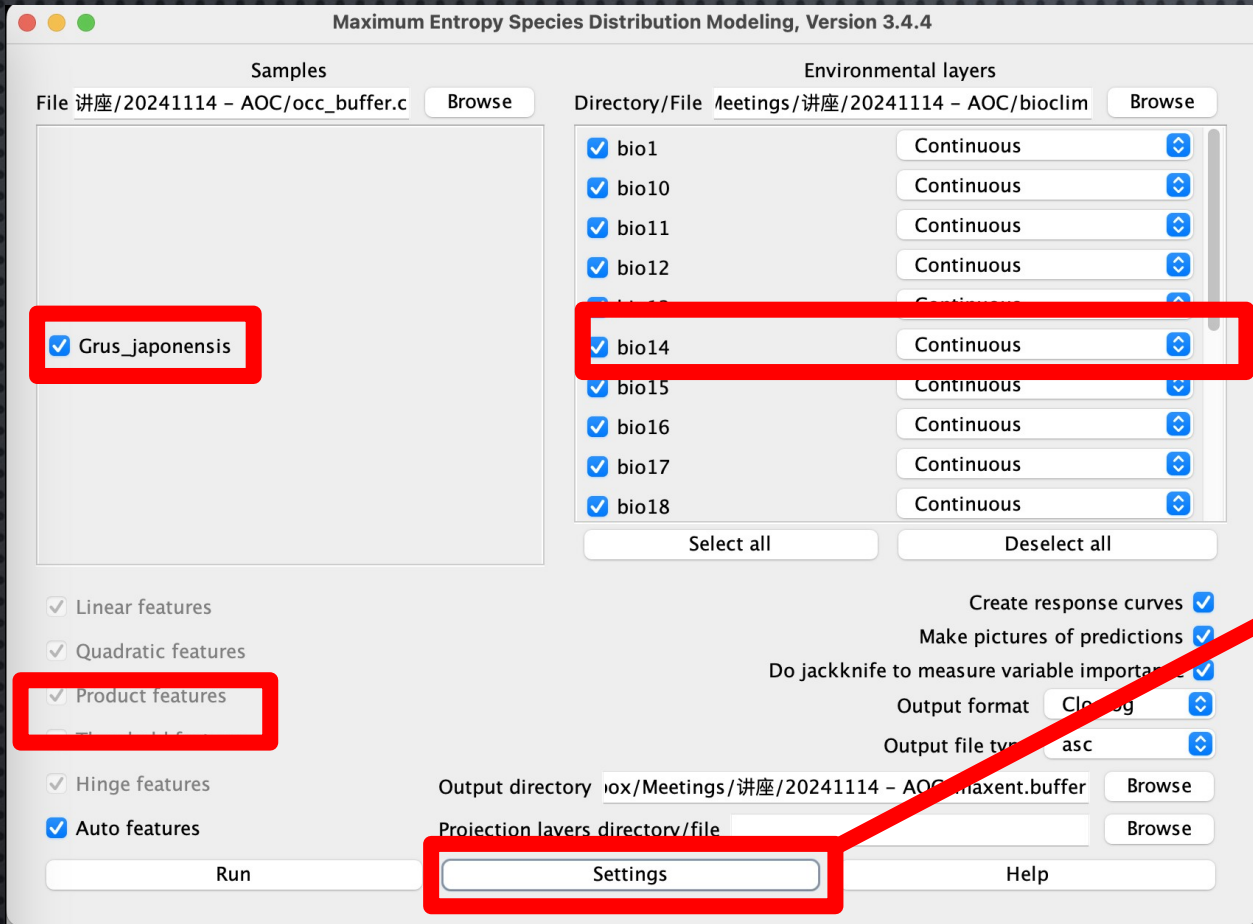
新的 MAXENT 模型

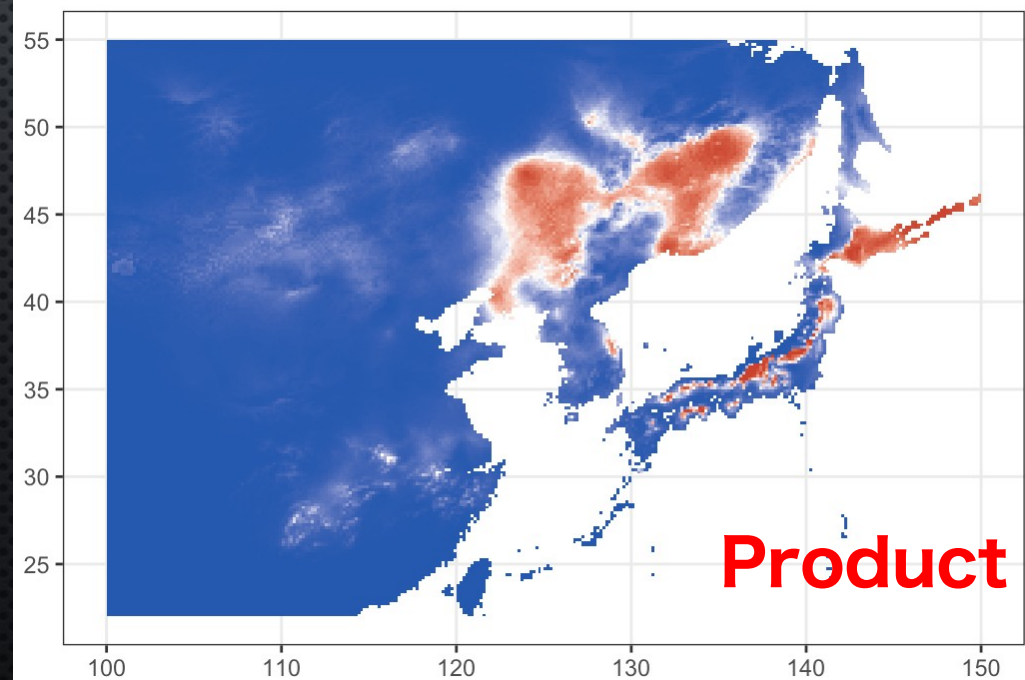
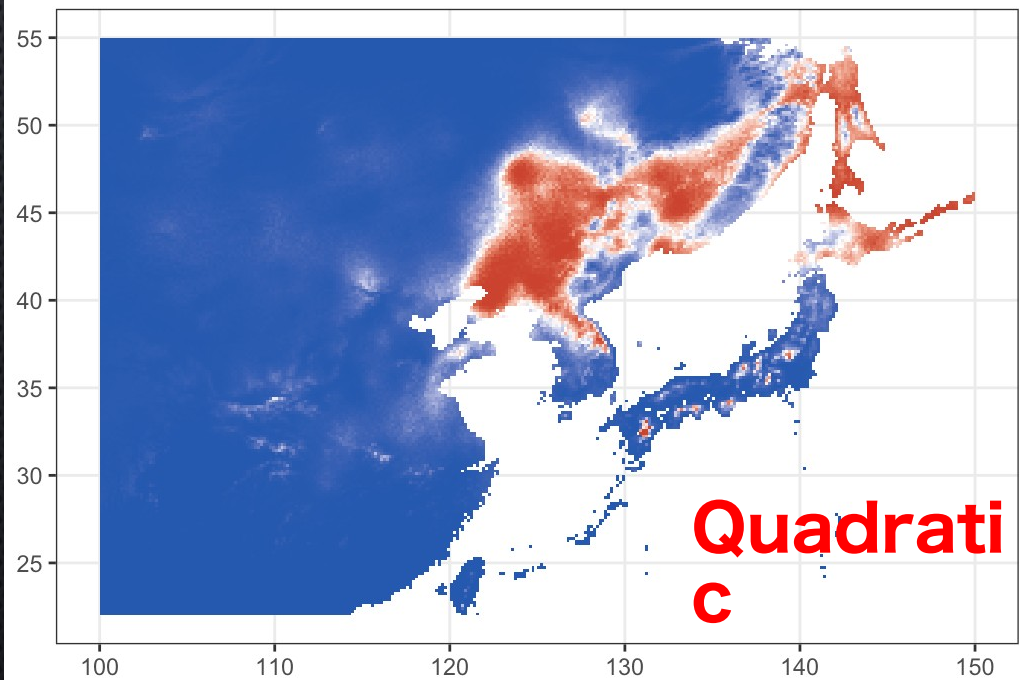
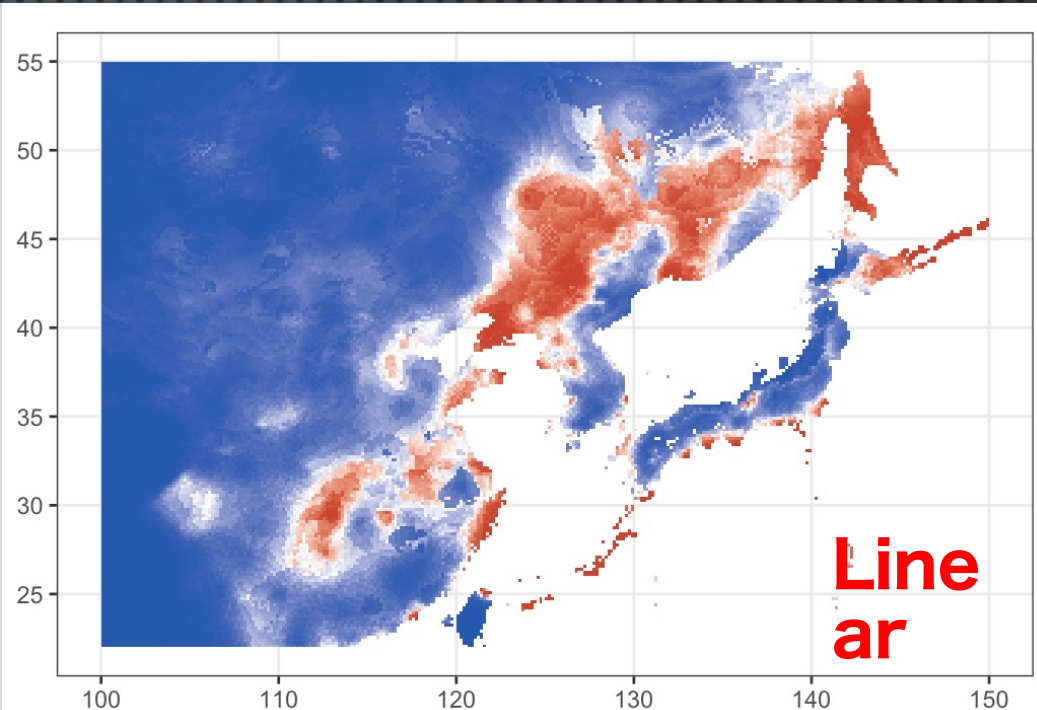
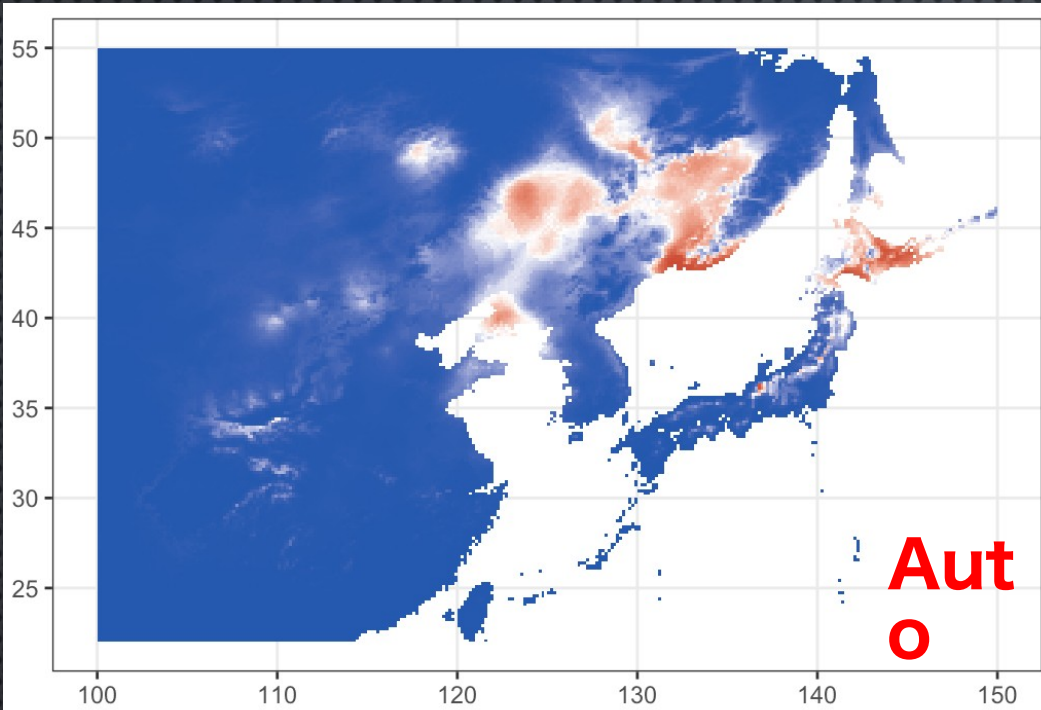


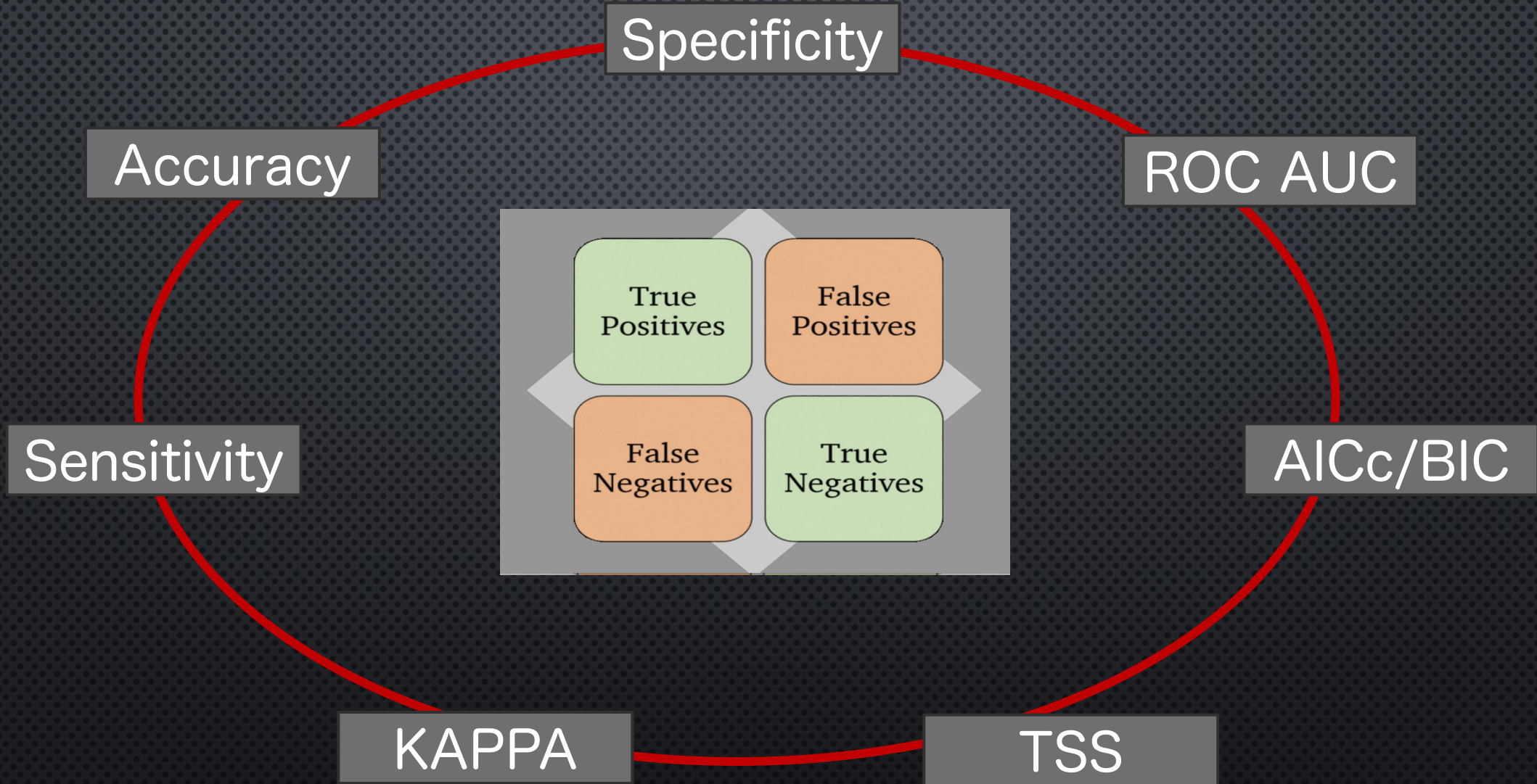
这就够了吗？

NO!









模型评估

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
Predicted condition	Predicted condition positive	True positive , Power	False positive , Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$
	Predicted condition negative	False negative , Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection = $\frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ $F_1 \text{ score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

sensitivity, recall, hit rate, or true positive rate (TPR)

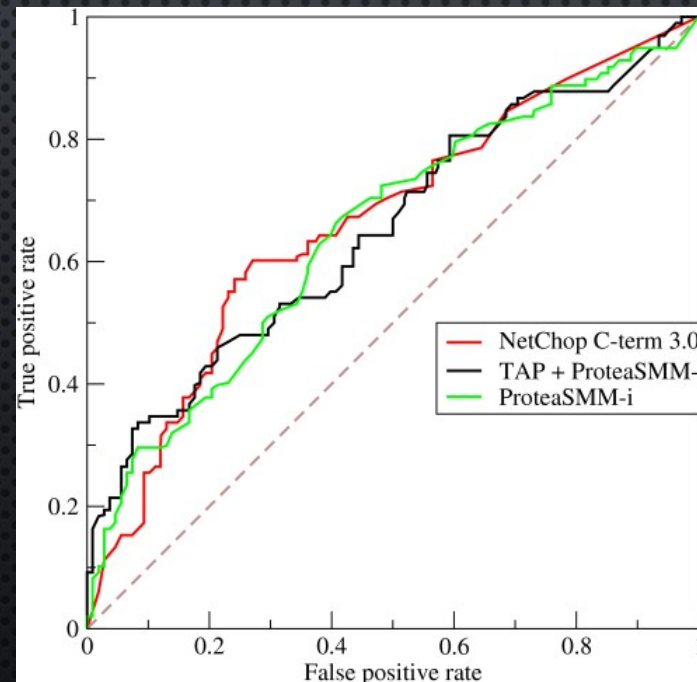
$$\text{TPR} = \frac{\text{TP}}{P} = \frac{\text{TP}}{\text{TP} + \text{FN}} = 1 - \text{FNR}$$

specificity, selectivity or true negative rate (TNR)

$$\text{TNR} = \frac{\text{TN}}{N} = \frac{\text{TN}}{\text{TN} + \text{FP}} = 1 - \text{FPR}$$

accuracy (ACC)

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{P + N} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$



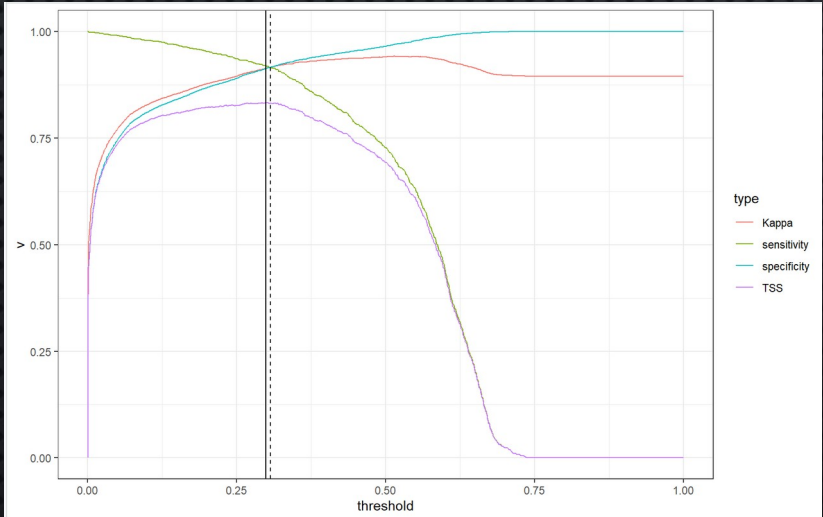
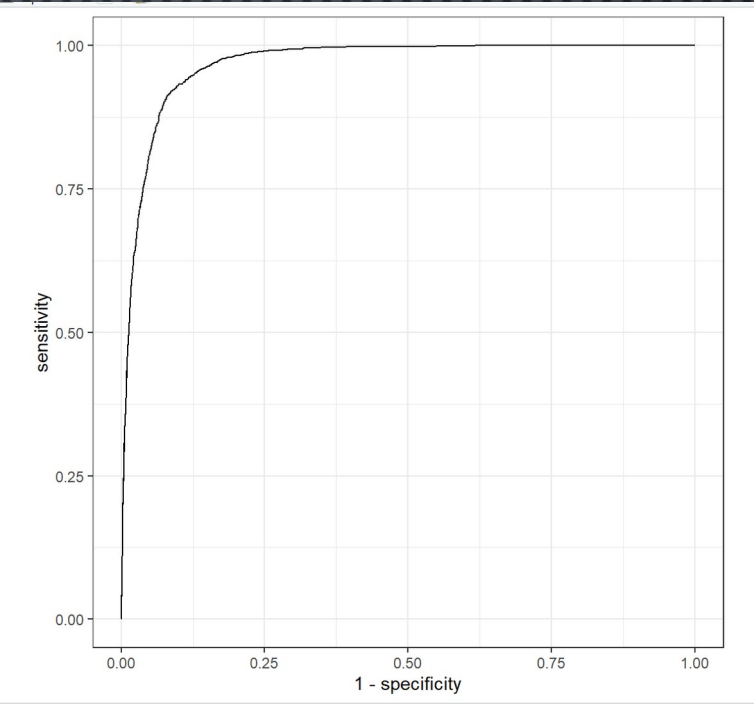
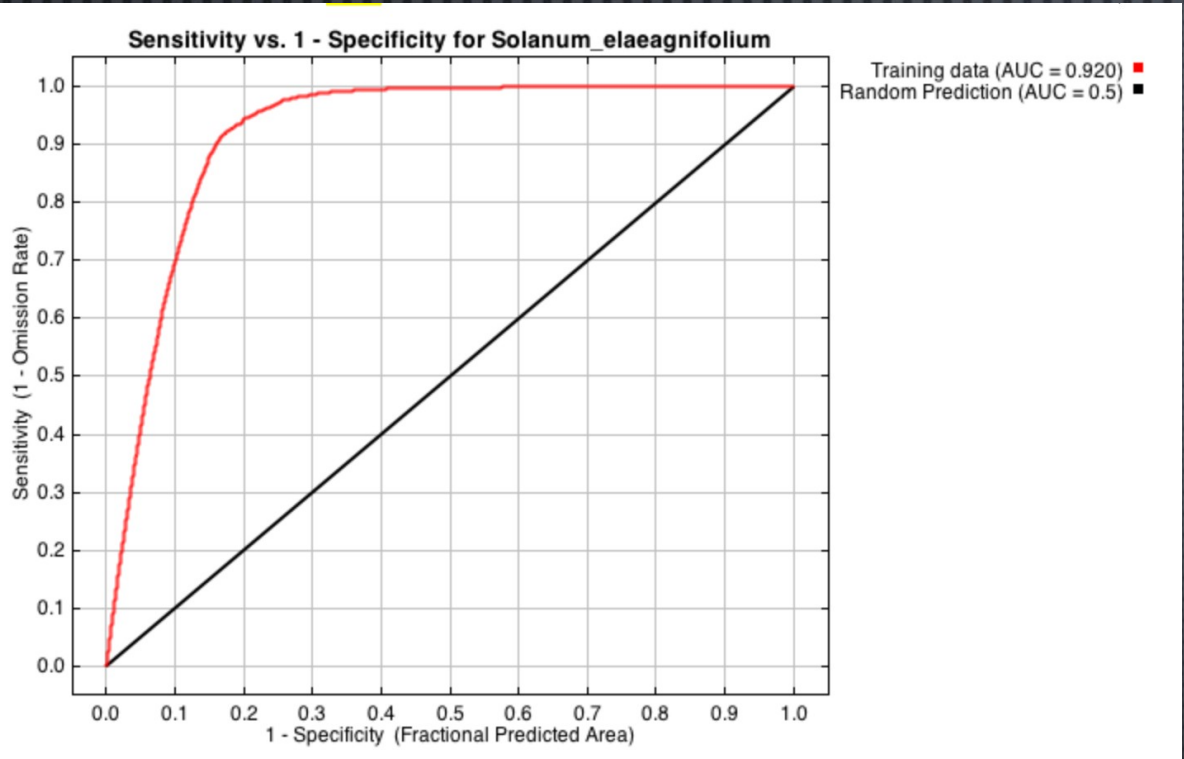
模型评估

		B	
		Yes	No
A	Yes	a	b
	No	c	d

- Kappa vs TSS

- $Kappa = \frac{a+d}{a+b+c+d}$

- $TSS = \frac{ad-bc}{(a+c)(b+d)} = Sensitivity + Specificity - 1$



Cumulative threshold	Logistic threshold	Description	Fractional predicted area	Training omission rate
1.000	0.027	Fixed cumulative value 1	0.379	0.006
5.000	0.144	Fixed cumulative value 5	0.247	0.030
10.000	0.248	Fixed cumulative value 10	0.197	0.063
0.092	0.002	Minimum training presence	0.601	0.000
16.203	0.333	10 percentile training presence	0.160	0.100
19.829	0.379	Equal training sensitivity and specificity	0.144	0.144
13.187	0.294	Maximum training sensitivity plus specificity	0.176	0.077
2.143	0.060	Balance training omission, predicted area and threshold value	0.315	0.010
5.265	0.152	Equate entropy of thresholded and original distributions	0.244	0.032

ENMeval version 2.0.4

R package for automated tuning and evaluations of ecological niche models

NOTE: ENMeval is a work in progress, changing slowly to fix bugs when users identify them. If you find a bug, please raise an Issue in this Github repo and I will resolve it as soon as I can. The CRAN version may lag behind the Github one, so please try the development version here first if you are having any issues. Install with:

```
devtools::install_packages("jamiemkass/ENMeval")
```

[ENMeval](#) is an R package that performs automated tuning and evaluations of ecological niche models and species distribution models. Version >=2.0.0 represents an extensive restructure and expansion of version 0.3.1, and has many new features, including customizable specification of algorithms besides Maxent using the new [ENMdetails](#) object, comprehensive metadata output, null model evaluations, new visualization tools, a completely updated and extensive [vignette](#) with a complete analysis walkthrough, and more flexibility for different analyses and data types. Many of these new features were created in response to user requests -- thank you for your input!

[ENMeval](#) >=2.0.0 includes the functionality to specify any algorithm of choice, but comes out of the box with two implementations of Maxent: maxnet ([Phillips et al. 2017](#)) from the [maxnet R package](#) and the Java software maxent.jar ([Phillips et al. 2006](#)), available [here](#), as well as BIOCLIM implemented with the [dismo R package](#).

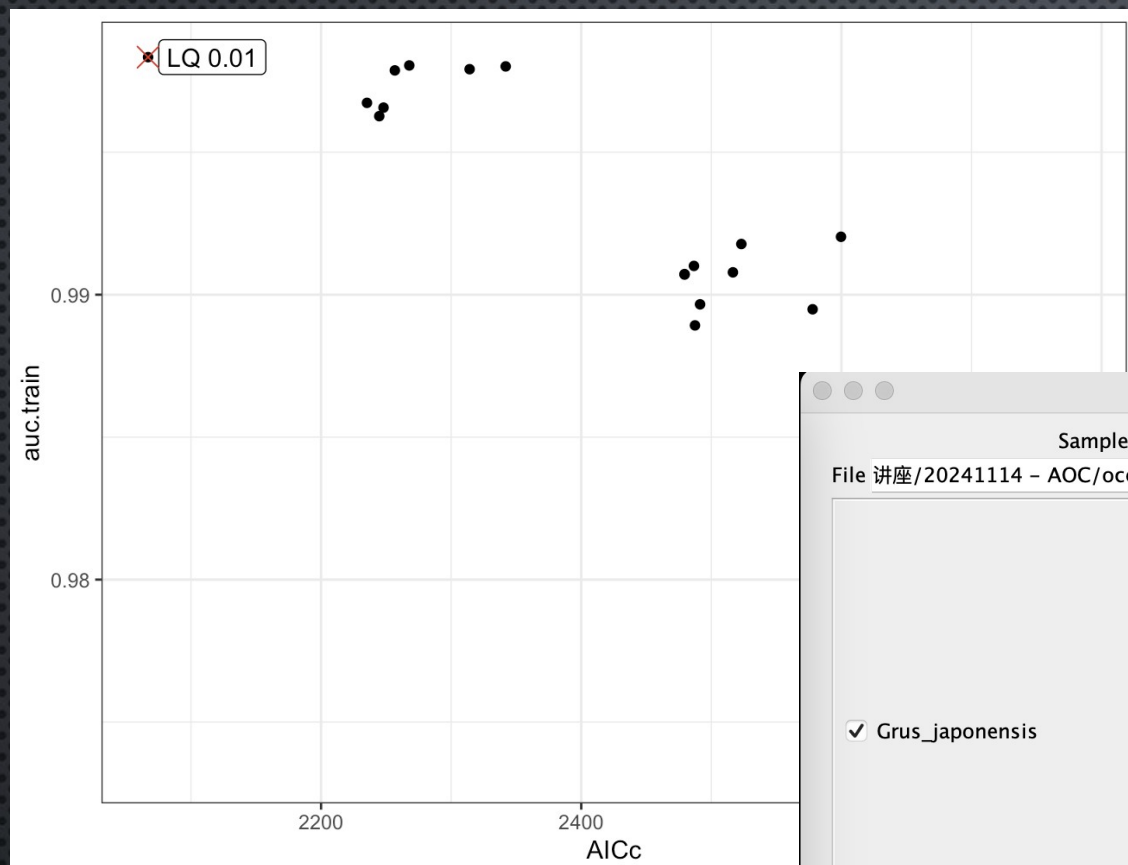
Model tuning refers to the process of building models with varying complexity settings, then choosing optimal settings based on some criteria. As it is difficult to predict in advance what level of complexity best fits your data and results in the most ecologically realistic response for your species, model tuning and evaluations are essential for ENM studies. This process helps researchers maximize predictive ability and avoid overfitting with models that are too complex.

```
bccs_ensembleval<-df[[is_in_buffer==1, c("x", "y")]]
envs.files<-list.files('bioclim', pattern="\\.asc", full.names = T)

v<-data.table(raster::extract(envs, bccs_ensembleval))
v$sum<-rowSums(v)
v[is.na(sum)]
v[is.na(v)]
envs <- raster::stack(envs.files)
ensembleval_result<-ENMevaluate(occs = bccs_ensembleval, envs = envs, bg = NULL,
                                algorithm = 'maxent.jar', partitions = 'block',
                                tune.args =
                                  list(fc = c("L", "Q", "P", "LQ", "QP"),
                                       rm = c(0.01, 0.1, 1, 10)))

ensembleval_elu<-data.table(ensembleval_result@results)
saveRDS(ensembleval_elu, "ensembleval_elu.rda")
ensembleval_elu<-readRDS("ensembleval_elu.rda")
ensembleval_elu$label<-paste(ensembleval_elu$fc, ensembleval_elu$rm, sep=" ")
best<-ensembleval_elu[AICc==min(ensembleval_elu$AICc) | auc.train==max(ensembleval_elu$auc.train)]
ggplot(ensembleval_elu)+
  geom_point(aes(x=AICc, y=auc.train))+
  geom_label_repel(data=best, aes(x=AICc, y=auc.train, label=label))+
  geom_point(data=best,
            aes(x=AICc, y=auc.train), color="#DC3220", shape=4, size=4)+
  theme_bw()
```

	▲ fc ↕	rm ↕	tune.args ↕	auc.train ↕	cbi.train ↕	auc.diff.avg ↕	auc.diff.sd ↕	auc.val.avg ↕	auc.val.sd ↕	c
1	L	0.01	fc.L_rm.0.01	0.9920320	NA	0.010948597	0.014633828	0.9736610	0.021167078	
2	Q	0.01	fc.Q_rm.0.01	0.9907844	NA	0.020329581	0.029367181	0.9707149	0.034205465	
3	P	0.01	fc.P_rm.0.01	0.9980123	NA	0.010158314	0.015559521	0.9842386	0.015604793	
4	LQ	0.01	fc.LQ_rm.0.01	0.9983369	NA	0.003765547	0.003189902	0.9949495	0.004076814	
5	QP	0.01	fc.QP_rm.0.01	0.9980459	NA	0.006584345	0.008360807	0.9884176	0.009539667	
6	L	0.1	fc.L_rm.0.1	0.9917787	NA	0.012148530	0.017700941	0.9694382	0.026254008	
7	Q	0.1	fc.Q_rm.0.1	0.9910098	NA	0.024065911	0.034859661	0.9676135	0.039356661	
8	P	0.1	fc.P_rm.0.1	0.9979139	NA	0.012359335	0.021271479	0.9832085	0.020548954	
9	LQ	0.1	fc.LQ_rm.0.1	0.9979615	NA	0.004628552	0.006250168	0.9929173	0.007439913	
10	QP	0.1	fc.QP_rm.0.1	0.9978721	NA	0.009343484	0.012720096	0.9859263	0.012760532	
11	L	1	fc.L_rm.1	0.9896615	NA	0.016926970	0.027257180	0.9632502	0.032623004	
12	Q	1	fc.Q_rm.1	0.9889221	NA	0.024438036	0.031428740	0.9643835	0.040956881	
13	P	1	fc.P_rm.1	0.9967328	NA	0.016749436	0.028714662	0.9767716	0.029368801	
14	LQ	1	fc.LQ_rm.1	0.9962656	NA	0.018185850	0.028480532	0.9737189	0.031553472	
15	QP	1	fc.QP_rm.1	0.9965656	NA	0.017599595	0.030058168	0.9758655	0.030726120	
16	L	10	fc.L_rm.10	0.9768484	NA	0.041563116	0.066383945	0.9205814	0.084034189	
17	Q	10	fc.Q_rm.10	0.9733926	NA	0.022882920	0.020489621	0.9543310	0.030080706	



Maximum Entropy Species Distribution Modeling, Version 3.4.4

Samples
File 讲座/20241114 - AOC/occ_buffer.c Browse

Environmental layers
Directory/File meetings/讲座/20241114 - AOC/bioclim Browse

Maximum Entropy Parameters

Basic Advanced Experimental

- ☐ Random seed
- ☒ Give visual warnings
- ☒ Show tooltips
- ☒ Ask before overwriting
- ☐ Skip if output exists
- ☒ Remove duplicate presence records
- ☒ Write clamp grid when projecting
- ☒ Do MESS analysis when projecting

Random test percentage

Regularization multiplier

Max number of background points

Replicates

Replicated run type

Test sample file Browse

Output directory

Projection layer

Run Settings Help

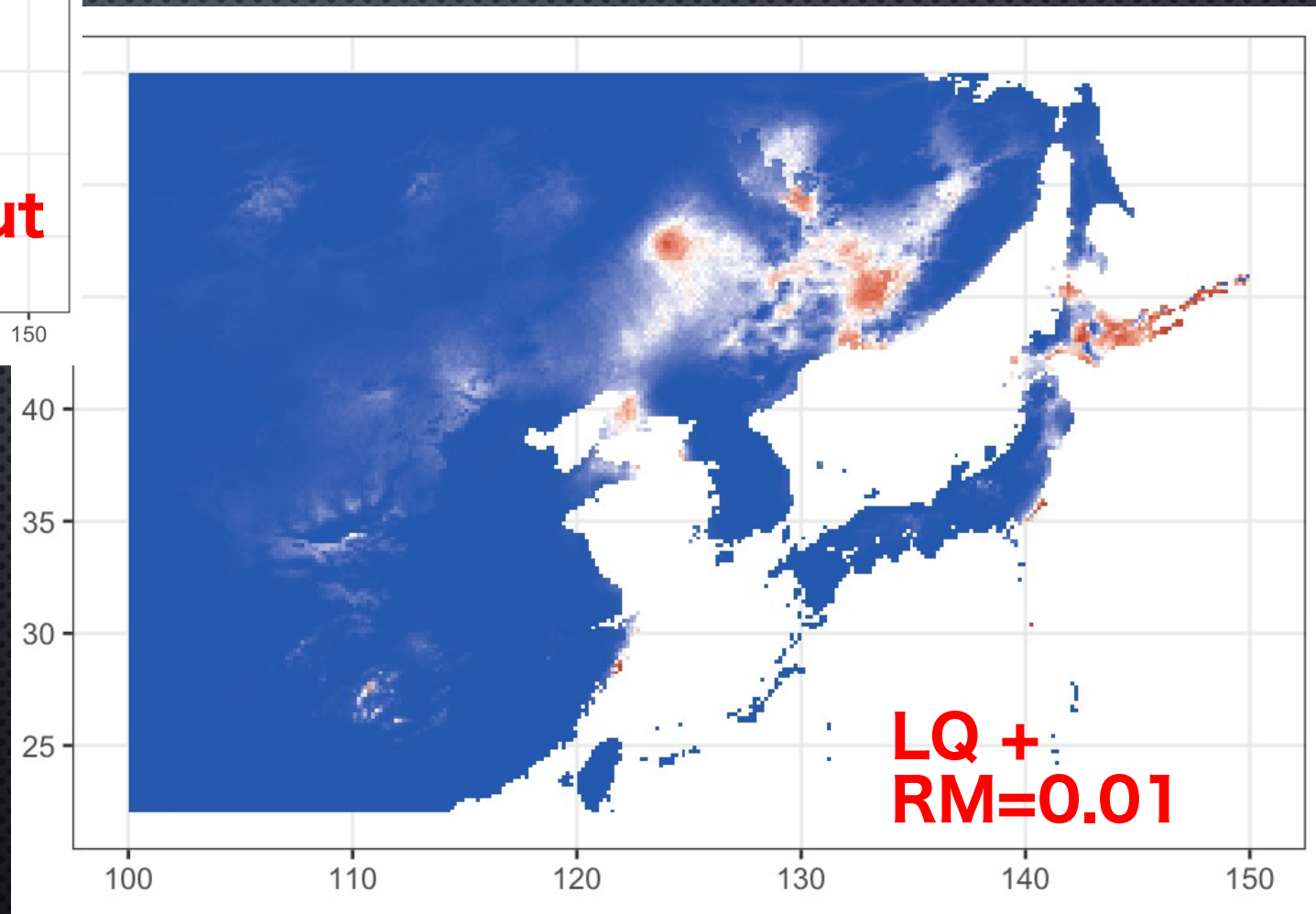
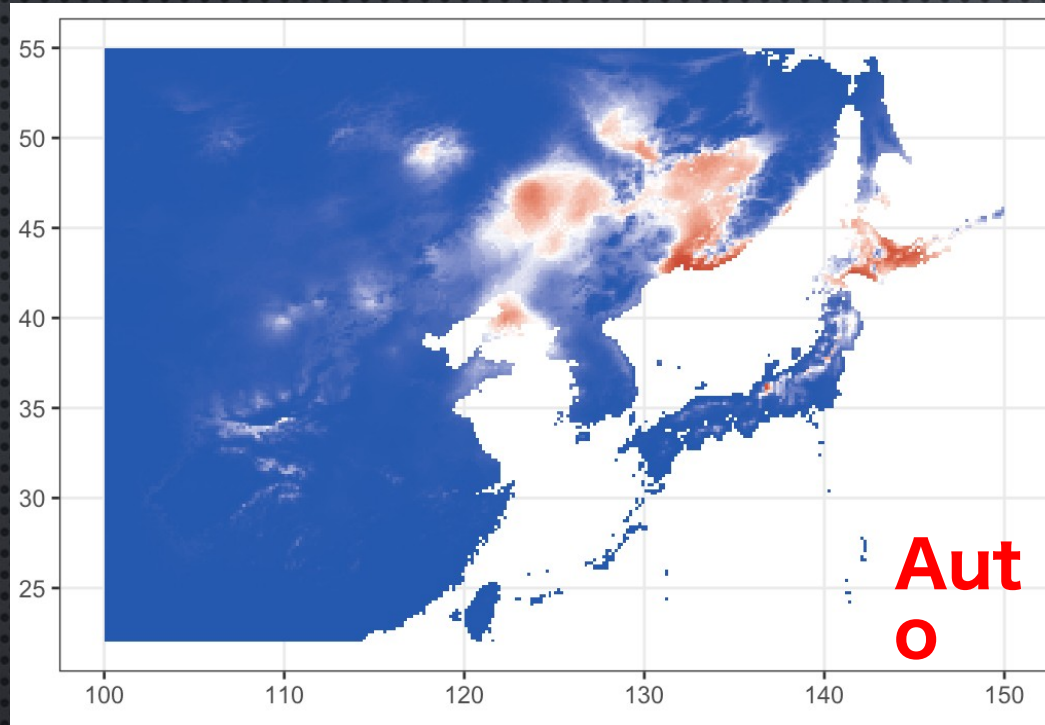
☒ Linear features

☐ Product features

☐ Threshold features

☐ Hinge features

☐ Auto features



那么我们得到最好的模型了吗？



All models are wrong, but some are
useful.

— *George E. P. Box* —

AZ QUOTES

接下来还需要做些什么？

- 更多的生态位模型
- 环境变量选择
- 如何生成伪负样本 / 背景点
- 缓冲区
- 等等.....

流行的生态位模型 / 物种分布模型

Model	Algorithm	Key Features
Maxent	Maximum Entropy	- Presence-only data - Robust and user-friendly - Handles complex interactions well
GLM	Generalized Linear Model	- Regression-based - Models linear relationships - Flexible and interpretable
GAM	Generalized Additive Model	- Nonlinear extensions of GLM - Captures complex environmental interactions
BRT	Boosted Regression Trees	- Ensemble of regression trees - Handles complex and large datasets - High predictive power
Random Forest	Ensemble Decision Trees	- Builds multiple decision trees - Effective for nonlinear and high-dimensional data
ENFA	Ecological Niche Factor Analysis	- Presence-only data - Calculates niche in environmental space - Good for limited data
BIOCLIM	Bioclimatic Envelope Model	- Bioclimatic variables - Simple and interpretable - Best for preliminary analyses
GARP	Genetic Algorithm	- Uses genetic algorithms - Builds rule sets from environmental variables
MARS	Multivariate Adaptive Regression Splines	- Flexible regression model - Handles nonlinear interactions - Suitable for small datasets
CLIMEX	Mechanistic Model	- Uses climatic and physiological data - Often applied in agriculture and invasion studies
SDM Frameworks	Various (e.g., Biomod2, Wallace, dismo)	- Combines multiple SDMs - Ensemble modeling - Increases robustness and accuracy

环境变量

BIO1 = ANNUAL MEAN TEMPERATURE

BIO2 = MEAN DIURNAL RANGE (MEAN OF MONTHLY (MAX TEMP - MIN TEMP))

BIO3 = ISOTHERMALITY (BIO2/BIO7) ($\times 100$)

BIO4 = TEMPERATURE SEASONALITY (STANDARD DEVIATION $\times 100$)

BIO5 = MAX TEMPERATURE OF WARMEST MONTH

BIO6 = MIN TEMPERATURE OF COLDEST MONTH

BIO7 = TEMPERATURE ANNUAL RANGE (BIO5-BIO6)

BIO8 = MEAN TEMPERATURE OF WETTEST QUARTER

BIO9 = MEAN TEMPERATURE OF DRIEST QUARTER

BIO10 = MEAN TEMPERATURE OF WARMEST QUARTER

BIO11 = MEAN TEMPERATURE OF COLDEST QUARTER

BIO12 = ANNUAL PRECIPITATION

BIO13 = PRECIPITATION OF WETTEST MONTH

BIO14 = PRECIPITATION OF DRIEST MONTH

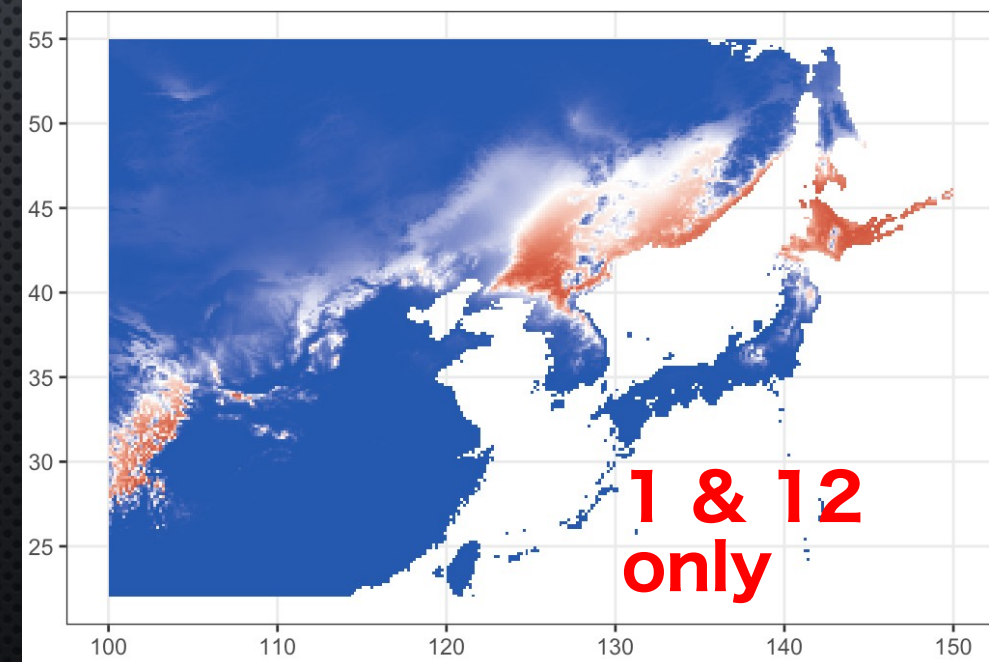
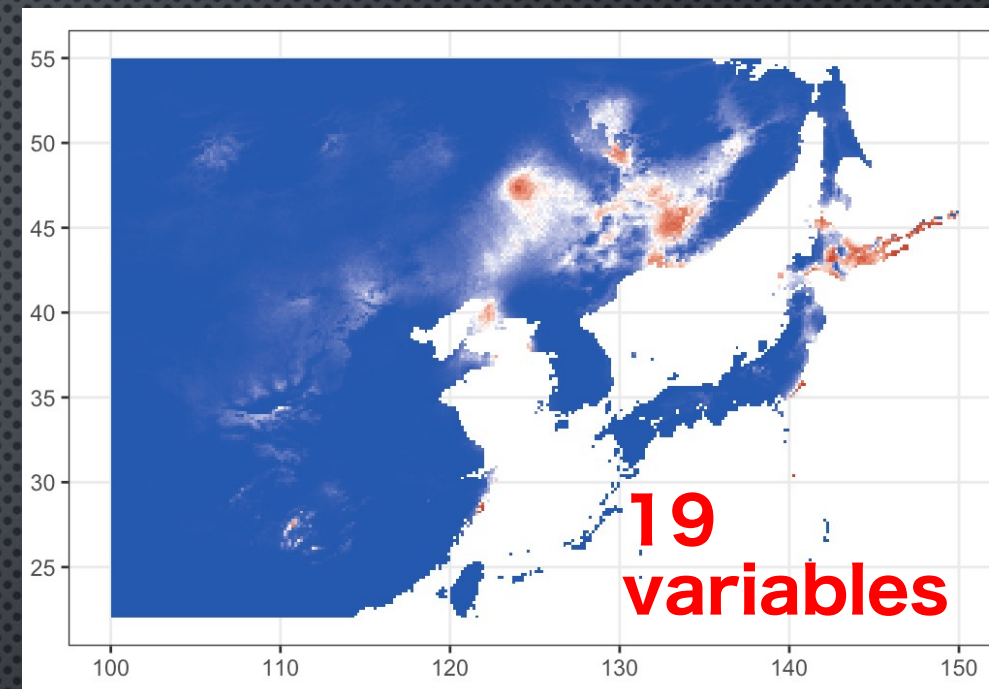
BIO15 = PRECIPITATION SEASONALITY (COEFFICIENT OF VARIATION)

BIO16 = PRECIPITATION OF WETTEST QUARTER

BIO17 = PRECIPITATION OF DRIEST QUARTER

BIO18 = PRECIPITATION OF WARMEST QUARTER

BIO19 = PRECIPITATION OF COLDEST QUARTER



背景点数目

Maximum Entropy Parameters

Basic Advanced Experimental

- ☐ Random seed
- ☒ Give visual warnings
- ☒ Show tooltips
- ☒ Ask before overwriting
- ☐ Skip if output exists
- ☒ Remove duplicate presence records
- ☒ Write clamp grid when projecting
- ☒ Do MESS analysis when projecting

Random test percentage

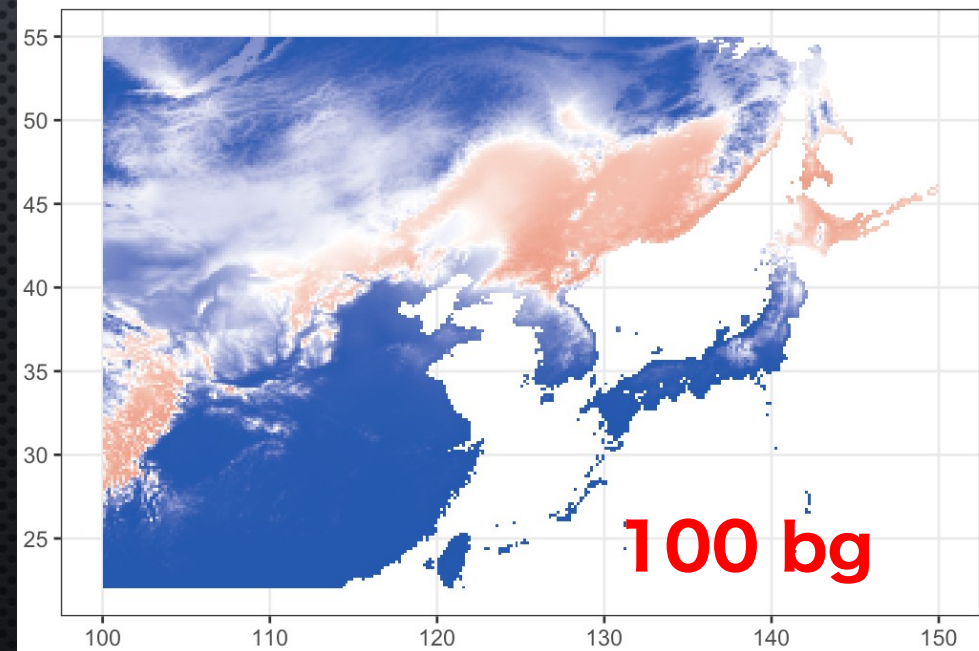
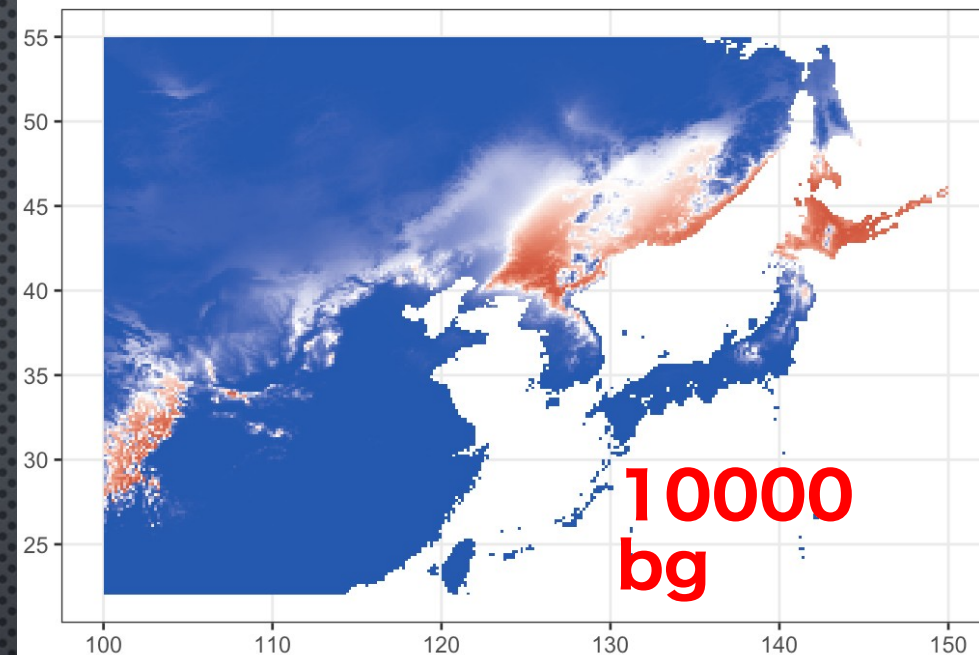
Regularization multiplier

Max number of background points

Replicates

Replicated run type

Test sample file



核心信息

- 生态位模型概念上简单，但在实践中复杂
- 生态位模型不仅限于 Maxent 模型
- 不存在万能解决方案
- 生态学洞察力至关重要
- 数据是核心
- 生态位模型只是生态研究中的一个工具
- 谨慎选择环境变量
- 模型评估至关重要
- 推荐使用 R 语言进行生态位模型分析